

Name: _____



SCEGGS Darlinghurst
HSC ASSESSMENT 21 May 2001

Mathematics

Year 12

Extension 1

TIME ALLOWED: $1\frac{1}{2}$ HOURS

Topics: *Permutations and Combinations, Inverse Functions, Further Trigonometry, Binomial Probability.*

Percentage of final assessment: 35%

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- Ensure that your name is on this booklet.
- All questions are of equal value.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Approved calculators should be used. Mathematical templates and geometrical instruments may be used.
- Begin each question on a new page - write your name at the top of each page.
- Hand up the question sheet as well as your answer pages.

QUESTION 1

(18 Marks)

Marks

a) Evaluate $\int_1^{\sqrt{2}} \frac{dx}{\sqrt{4-x^2}}$

3

b) The probability that a machine fails in repeated trials is 0.05. If the machine is operated 10 times, find the probability that:

(i) it does not fail.

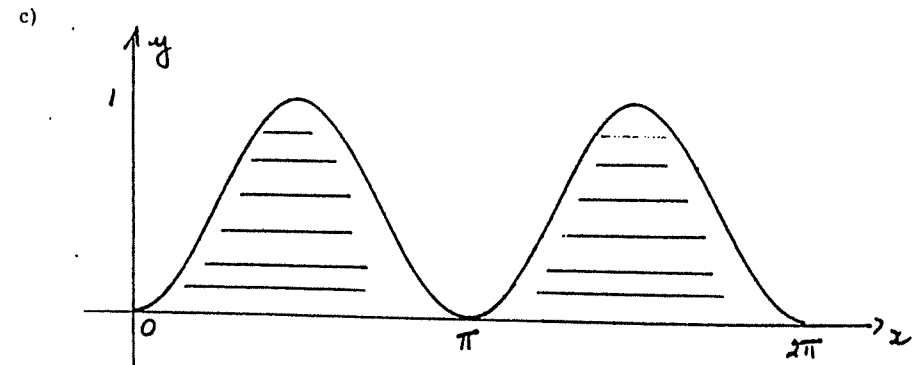
1

(ii) it fails more than once

2

(iii) the company that makes the machine states that "the machine has a 99% certainty that it will not fail more than once in 100 starts." How accurate is this statement? Explain your answer.

2



The graph shown above is $y = \sin^3 x$ for $0 \leq x \leq 2\pi$

Find the shaded area.

3

- d.) There are six children in a group, sitting in a circle.
- In how many ways can they sit if they may sit anywhere?
 - In how many ways can they sit if two particular students are not allowed to sit together?
 - Three of the children are sisters. Find the probability that they sit together.
 - If the group contains three boys and three girls, find the probability that the boys and girls are seated alternately.

Marks

1

2

2

2

QUESTION 2

(20 Marks)

START A NEW PAGE

- a.) (i) Prove that $\cos^2 2x = \frac{1}{2}(\cos 4x + 1)$
- (ii) Hence find the volume formed when the region contained by the x axis and the curve $y = \cos 2x$ between $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$ is rotated about the x axis. (Answer in exact form.)
- b.) A tin contains a large number of red and green lollies mixed together in the ratio 3:2. A handful of 20 lollies is taken from the tin. Find the probability that:
- there are no red lollies in the handful.
 - there are 7 red lollies in the handful.
 - there are at least 18 red lollies.
 - the handful is such that it can be shared between 5 children so that each child receives the same number of red lollies. (There is no need to calculate this answer.)

1

4

1

1

2

3

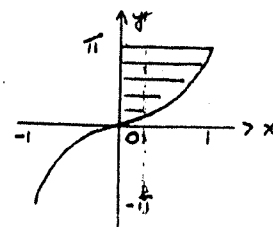
Marks

- c.) Find the exact value of:

$$\sin \left(\sin^{-1} \left(\frac{4}{5} \right) - \tan^{-1} \left(\frac{-5}{12} \right) \right)$$

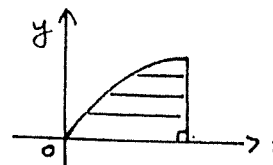
3

- d.)



The curve shown is:
 $y = 2 \sin^{-1} x$

A region is shaded



- (i)

The curve shown is a sine curve.
The region shaded has the same area as that shown above.

2

State the equation of the curve and the domain and range of the shaded region.

- (ii) Hence find the volume formed when $y = 2 \sin^{-1} x$ between $y = 0$ and $y = \pi$ is rotated about the y axis.

3

QUESTION 3 (17 Marks)

START A NEW PAGE

Marks

- a) (i) In how many ways can a committee of 6 people be formed from 5 women and 4 men? 1
- (ii) How many committees are possible if a particular man and a particular woman refuse to serve together? 2
- (iii) If one of the committees is chosen at random, find the probability that it contains more than 3 women. 3
- b) (i) How many different arrangements are possible of the word **DELETED**? 1
- (ii) In how many of these arrangements are the E's together? 2
- (iii) If six of the letters are selected at random, how many different arrangements are now possible? 3
- c) 4 married couples play 2 games of tennis doubles at the same time. How many ways can the pairings be arranged:
- (i) if there are no restrictions? 3
- (ii) the married couples play together? 2

QUESTION 4 (17 Marks)

START A NEW PAGE

Marks

- a) Consider the function $y = \frac{1}{2} \cos^{-1} \frac{x}{2}$
- (i) State the domain and range of the function. 2
- (ii) Sketch the function. 2
- b) Evaluate $\int_{-\frac{2}{3}}^{\frac{2}{3}} \frac{dx}{4 + 9x^2}$ 4
- c) (i) Sketch $f(x) = x^2 + 2x + 1$ 1
- (ii) State a suitable domain (containing $x = 0$) for $f(x)$ such that $f(x)$ has an inverse function. 1
- (iii) Sketch the inverse function $f^{-1}(x)$ showing all important features. 2
- (iv) Find $f^{-1}(x)$
- d) Consider $y = \cos^{-1}(\sin x)$
- (i) Find $\frac{dy}{dx}$ 1
- (ii) State an interesting fact about $\frac{dy}{dx}$ 1
- (iii) Sketch $y = \cos^{-1}(\sin x)$ 3

Assessment Test Answers Extension 1 Test 2.

$$\begin{aligned} \text{a) } \int_1^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}} &= \left[\sin^{-1} \frac{x}{2} \right]_1^{\sqrt{3}} \\ &= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2} \\ &= \frac{\pi}{3} - \frac{\pi}{6} \\ &= \frac{\pi}{6} \end{aligned}$$

$$\text{b) (i) } (0.95)^{10} = 0.5987 \dots$$

$$\begin{aligned} \text{(ii) Prob (0,1 failures)} &= (0.95)^{10} + 10 \times (0.95)^9 \times (0.05) \\ &= 0.914 \dots \end{aligned}$$

$$\begin{aligned} \therefore \text{Prob (more than 1 failure)} &= 1 - 0.914 \dots \\ &= 0.086 \dots \end{aligned}$$

$$\begin{aligned} \text{(iii) Prob (not failing more than one)} &= 100 \text{ successes} + 99 \text{ successes} \\ &= (0.95)^{100} + 100 \times (0.95)^{99} \times (0.05) \\ &= 0.037 \dots \end{aligned}$$

statement is extremely inaccurate.

$$\begin{aligned} \text{c) Area} &= 2 \int_0^{\pi} \sin^2 x \, dx \\ &= 2 \int_0^{\pi} \frac{1 - \cos 2x}{2} \, dx \\ &= \left[x - \frac{\sin 2x}{2} \right]_0^{\pi} \\ &= \pi - \frac{\sin 2\pi}{2} - 0 + \frac{\sin 0}{2} \\ &= \pi \, \text{m}^2 \end{aligned}$$

$$\text{d) (i) } 5! = 120$$

$$\begin{aligned} \text{(ii) if together} &= 4! \times 2 = 48 \\ \text{if not together} &= 120 - 48 = 72 \end{aligned}$$

$$\begin{aligned} \text{(iii) if together} &= 3! \times 3! = 36 \\ \therefore \text{probability} &= \frac{36}{120} = \frac{3}{10} \end{aligned}$$

$$\begin{aligned} \text{(iv) if alternating} &= 3! \times 2! = 12 \\ \therefore \text{probability} &= \frac{12}{120} = \frac{1}{10} \end{aligned}$$

$$\begin{aligned} \text{(2) a) (i) } \cos 4x &= \cos^2 2x - \sin^2 2x \\ &= 2\cos^2 2x - 1 \\ \therefore \cos^2 2x &= \frac{1}{2} (\cos 4x + 1) \end{aligned}$$

$$\begin{aligned} \text{(ii) } V &= \pi \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \cos^2 2x \, dx \\ &= \frac{\pi}{2} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} (\cos 4x + 1) \, dx \\ &= \frac{\pi}{2} \left[\frac{\sin 4x}{4} + x \right]_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \\ &= \frac{\pi}{2} \left(\frac{\sin 3\pi}{4} + \frac{3\pi}{4} - \frac{\sin \pi}{4} - \frac{\pi}{4} \right) \\ V &= \frac{\pi^2}{4} \, \text{m}^3 \end{aligned}$$

$$\text{b) (i) } P(\text{no red}) = \left(\frac{2}{5}\right)^{20} = 1.0995 \times 10^{-8}$$

$$\text{(ii) } P(7 \text{ red}) = \binom{20}{7} \left(\frac{3}{5}\right)^7 \left(\frac{2}{5}\right)^{13} = 0.015$$

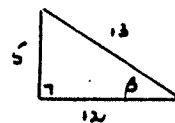
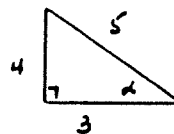
$$\begin{aligned} \text{(iii) } P(18, 19, 20 \text{ red}) &= \binom{20}{18} \left(\frac{3}{5}\right)^{18} \left(\frac{2}{5}\right)^2 + \binom{20}{19} \left(\frac{3}{5}\right)^{19} \left(\frac{2}{5}\right)^1 + \left(\frac{3}{5}\right)^{20} \\ &= 3.6 \times 10^{-3} \text{ (approx)} \end{aligned}$$

$$\begin{aligned} \text{(iv) The number of red lollies must be divisible by 5 i.e. } 0, 5, 10, 15, 20 \\ \left(\frac{2}{5}\right)^{20} + \binom{20}{5} \left(\frac{2}{5}\right)^5 \left(\frac{3}{5}\right)^{15} + \binom{20}{10} \left(\frac{2}{5}\right)^{10} \left(\frac{3}{5}\right)^{10} \\ + \binom{20}{15} \left(\frac{2}{5}\right)^{15} \left(\frac{3}{5}\right)^5 + \left(\frac{3}{5}\right)^{20} \end{aligned}$$

$$\text{c) Let } \sin^{-1}\left(\frac{4}{5}\right) = \alpha \quad (\sin \alpha = \frac{4}{5})$$

$$\tan^{-1}\left(-\frac{5}{12}\right) = \beta \quad (\tan \beta = -\frac{5}{12})$$

Note that α is 1st quadrant
 β is 4th quadrant



$$\begin{aligned} \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\ &= \frac{4}{5} \times \frac{12}{13} - \frac{3}{5} \times \frac{5}{13} \\ &= \frac{63}{65} \end{aligned}$$

$$\text{d) (i) } D: 0 \leq x \leq \pi$$

$$R: 0 \leq y \leq 1$$

$$\text{equation } y = \sin \frac{x}{2}$$

$$\begin{aligned} \text{(ii) } V &= \pi \int_0^{\pi} \sin^2 \frac{x}{2} \, dx \\ &= \frac{\pi}{2} \int_0^{\pi} 1 - \cos x \, dx \\ &= \frac{\pi}{2} \left[x - \sin x \right]_0^{\pi} \\ &= \frac{\pi}{2} \left[\pi - \sin \pi - 0 + \sin 0 \right] \\ V &= \frac{\pi^2}{2} \, \text{m}^3 \end{aligned}$$

$$\text{(3) a) (i) } \binom{9}{6} = 84$$

$$\begin{aligned} \text{(ii) if included} &= \binom{7}{4} = 35 \\ \therefore \text{if not together} &= 84 - 35 = 49 \end{aligned}$$

$$\text{(iii) 4 women} = \binom{5}{4} \times \binom{4}{2} = 30$$

$$\begin{aligned} 5 \text{ women} &= \binom{5}{5} \times \binom{4}{1} = 4 \\ \therefore \text{Probability} &= \frac{34}{84} = \frac{17}{42} \end{aligned}$$

$$\text{b) (i) } \frac{7!}{2! \times 3!} = 420$$

$$\text{(ii) } \frac{5!}{2!} = 60$$

$$\text{(iii) E excluded} = \frac{6!}{2! \times 2!} = 180$$

$$D \text{ excluded} = \frac{6!}{3!} = 120$$

$$L \text{ excluded} = \frac{6!}{3! \times 2!} = 60$$

$$T \text{ excluded} = \frac{6!}{3! \times 2!} = 60$$

$$\text{Total number} = 420$$

$$\text{c) (i) there are 2 methods either } \binom{8}{4} \div 2 \text{ to select 4 people}$$

$$\text{then } \binom{4}{2} \div 2 \text{ to organize the four into doubles pairings}$$

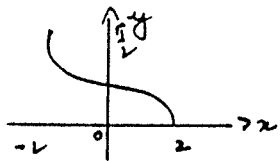
$$\therefore \binom{8}{4} \times \binom{4}{2} \times \binom{4}{2} \div 8 = 315$$

$$\text{or } \binom{8}{2} \times \binom{6}{2} \times \binom{4}{2} \div 8 = 315$$

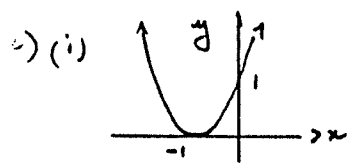
$$\text{(ii) } \binom{4}{2} \div 2 = 3$$

The division by 2 is the number of the games being played together

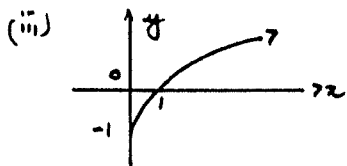
4) a) (i) $D: -2 \leq x \leq 2$
 $R: 0 \leq y \leq \frac{\pi}{2}$



b) $\int_{-\frac{2}{3}}^{\frac{2}{3}} \frac{dx}{4+9x^2} = \frac{1}{9} \int_{-\frac{2}{3}}^{\frac{2}{3}} \frac{dx}{\frac{4}{9} + x^2}$
 $= \frac{1}{9} \times \frac{3}{2} \left[\tan^{-1} \frac{3x}{2} \right]_{-\frac{2}{3}}^{\frac{2}{3}}$
 $= \frac{1}{6} \left[\tan^{-1}(1) - \tan^{-1}(-1) \right]$
 $= \frac{1}{6} \left(\frac{\pi}{4} + \frac{\pi}{4} \right)$
 $= \frac{\pi}{12}$



(ii) $x \geq -1$



(iv) $x = (y+1)^2$
 $y+1 = \pm \sqrt{x}$
 $y = -1 \pm \sqrt{x}$
 $f^{-1}(x) = -1 + \sqrt{x}$

d) $y = \cos^{-1}(\sin x)$

(i) $\frac{dy}{dx} = \frac{-1}{\sqrt{1-\sin^2 x}} \times \cos x$
 $= \frac{-\cos x}{\sqrt{\cos^2 x}}$

(ii) now $\sqrt{\cos^2 x} \geq 0$ by the definition of $\sqrt{\quad}$ but $\cos x$ can be +ve or -ve depending on the value of x
 $\therefore \frac{dy}{dx}$ can be 1 or -1

(iii) if $y = \cos^{-1}(\sin x)$

$-1 \leq \sin x \leq 1$ from the domain of an inverse cos.
 but this is true for all x
 $\therefore$ domain of $\cos^{-1}(\sin x)$ is all real x .

the range is $0 \leq y \leq \pi$
 Testing values gives the graph:

